

CSA1715-ARTIFICIAL INTELLIGENCE

LAB PROGRAM- 22

20-07-2026

22. WRITE A PROLOG PROGRAM TO PRINT PARTICIPANT BIRD CAN FLY OR NOT. INCORPORATE REQUIRED QUERIES.

AIM

To write and execute a Prolog program to determine whether a given bird can fly or cannot fly. The program uses facts and logical predicates to classify birds according to their ability to fly and demonstrates the use of queries in Prolog.

ALGORITHM

1. Start the program.
2. Define facts for birds that can fly.
3. Define facts for birds that cannot fly.
4. Define predicates can_fly/1 and cannot_fly/1.
5. Query the database with the name of a bird.
6. Check whether the bird belongs to the appropriate category.
7. Display the result.
8. Stop the program.

PROLOG PROGRAM

% Write a Prolog program to print whether a bird can fly or not.

can_fly(eagle).

can_fly(parrot).

can_fly(pigeon).

can_fly(sparrow).

`cannot_fly(penguin).`

`cannot_fly(ostrich).`

`cannot_fly(emu).`

SAMPLE INPUT

`?- can_fly(eagle).`

SAMPLE OUTPUT

`true.`

RESULT

Thus, the Prolog program to determine whether a bird can fly or not was successfully implemented and executed. The program correctly identified flying and non-flying birds using the facts stored in the knowledge base.